

Zijie Zhang

(586)306-1518 | zhangzijie1231@gmail.com | Los Angeles, CA 90025

Education

- University of Illinois Urbana-Champaign** | Urbana, IL 2021.08 - 2025.05
- Bachelor of Science in Computer Science & Statistics | 3.95 GPA | Phi Beta Kappa
- University of California, Los Angeles** | Los Angeles, CA 2025.09 - 2027.06
- Master of Science in Computer Science
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Experience

- Computer Vision and Application Development Intern** | Los Angeles Department of Transportation 2026.02 - Present
- Trained YOLOv8 models for traffic sign detection, improving recognition accuracy on labeled datasets and enabling scalable roadway asset recognition.
 - Developed data annotation pipelines using Label Studio + OCR, accelerating dataset preparation.
 - Designed a Google SSO workflow for an DOT account system, documenting authentication flows, implementation approach, user impact, and delivery-ready proposal materials.
 - Supported ADA compliance initiatives by helping review and improve web, mobile, and GIS-related content.
- Full-Stack Development Intern** | WOOLLY 2026.02 - Present
- Built voice-based training features in Next.js, including workplace scenario responses graded by OpenAI for professionalism, clarity, empathy, and task completion.
 - Implemented multilingual speech practice using Web Speech API, MediaRecorder fallback, OpenAI transcription, language-specific prompt banks, and locale-aware scoring.
- Full Stack Developer Intern** | Nanjing Yuanqi Medical Technology 2025.07 - Present
- Built a full-stack hospital maintenance management system (React, TypeScript, Node.js, Express, MySQL) with role-based access control, real-time dashboards, multi-format exports, and PM2-based deployment on internal server infrastructure.
 - Optimized backend queries using connection pooling, reducing response latency by ~30%.
 - Designed admin approval workflows and dynamic filtering/pagination for large datasets.
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Projects

- Change Impact Predictor: AST-Based API Change Mining & Test Impact Ranking** 2025.09 - 2025.12
- Built an AST-based system to detect API signature changes in Python repositories (pandas, scikit-learn).
 - Designed a scalable pipeline to map code changes to impacted test files using git-based analysis and multiprocessing.
 - Developed a ranking system evaluated with Recall@K and MRR to prioritize critical test cases.
- Yelp Review Simulation Agent (LLM-based)** 2025.09 - 2025.12
- Built an end-to-end LLM-driven user behavior simulation system for Yelp-style reviews, generating both star ratings and natural-language reviews from structured inputs, organized as a multi-stage pipeline with reproducible task selection and JSON logging.
 - Improved personalization with hybrid retrieval using Sentence-Transformers semantic similarity and recency weighting.
 - Validated impact through ablation modes comparing baseline against versions with planning, reasoning, memory, and grounding.
- 3GPP Intelligent Assistant & Profiler** 2025.09 - 2025.12
- Built a Full-Stack React Application featuring a hierarchical Clause History visualizer to track 3GPP document evolution.
 - Architected an Event-Driven RAG Pipeline: Designed a Python Middleware service using FastAPI that subscribes to backend events via an Event Bus pattern, enabling real-time status polling for the frontend and granular stage-wise latency tracking.
 - Implemented End-to-End System Profiling: Integrated tiktoken and custom metrology to measure token usage and latency.
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Skills

- **Languages:** Python, Java, JavaScript, TypeScript, C/C++, SQL
- **Frameworks:** React, Node.js, Express, FastAPI, Docker
- **ML/AI:** PyTorch, TensorFlow, scikit-learn, RAG, LLMs
- **Technical:** software development and testing, Linux, database management, data structures, data visualization, web development, and networking.